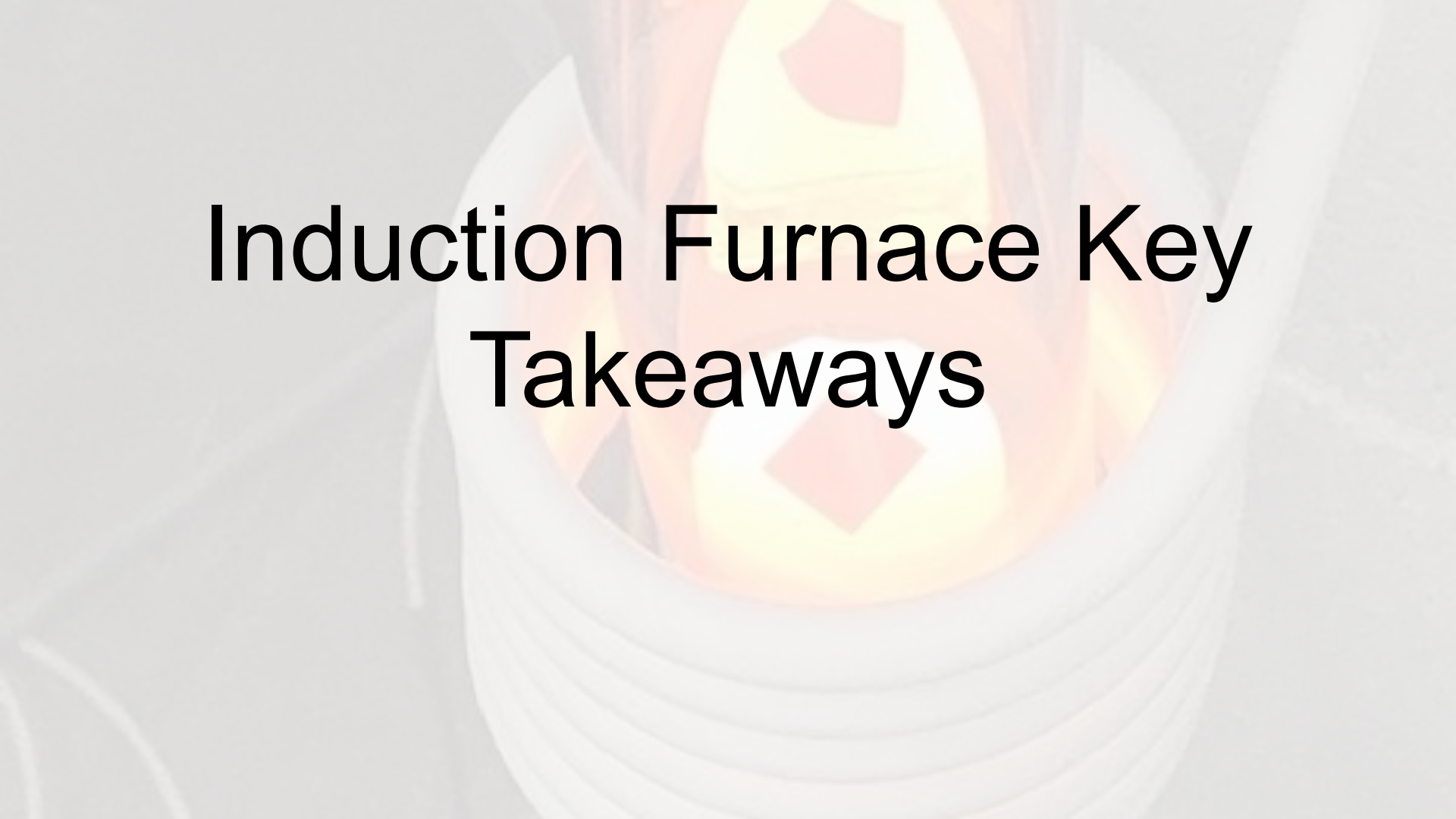
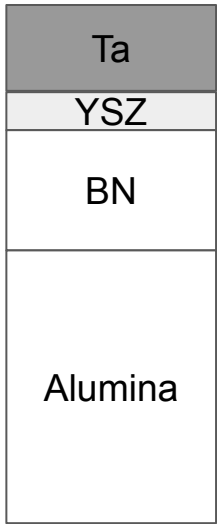




Induction Furnace Key Takeaways

Induction Furnace Configurations

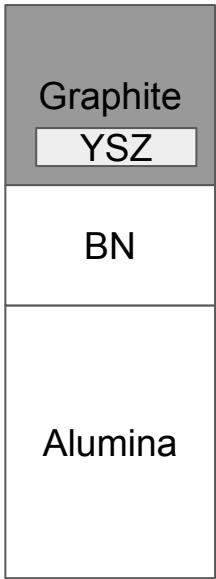
Shorter times and lower temperatures with all of these configurations have been successful at growing grains but the size was never satisfactory. (90 um clean samples achieved with setup #3 after 45 min. Anything longer than 3hrs contaminated sample)



2500C 8Hrs

Ta deposition on YSZ.
Otherwise successful

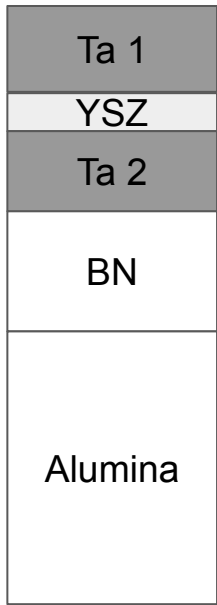
#1



2000C 168 Hrs

YSZ reacted with graphite.
YSZ Disappeared

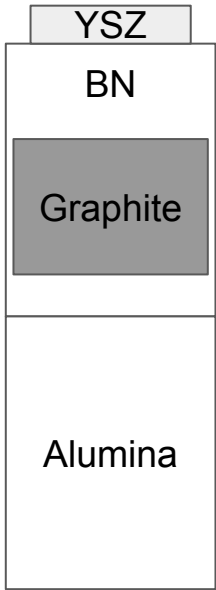
#2



2130C 80.4Hrs

Shell formed around Ta.
Hard to tell what happened.
Maybe BN reacted with Ta.

#3



Have not tried. We think this
could work. Graphite will be
contained in BN crucible. BN
and YSZ don't seem to react.

#4

Known Chemical Reactions

Alumina and Graphite

- Eutectic point around 1800C

Magnesium Oxide and Alumina

- Leaves strange black marks on MgO

Magnesium Oxide and Tantalum

- Diminishes the quality of the susceptor (Very little visual indicators)
- Oxidizes tantalum and melts

Carbon and YSZ (2000C)

- YSZ Vanishes

Oxygen and anything

YSZ and Boron Nitride (2500C)

- Changes the surface finish of the BN

Tantalum and Boron Nitride

- Changes surface color and finish and color of Ta

YSZ and Tantalum (2500C)

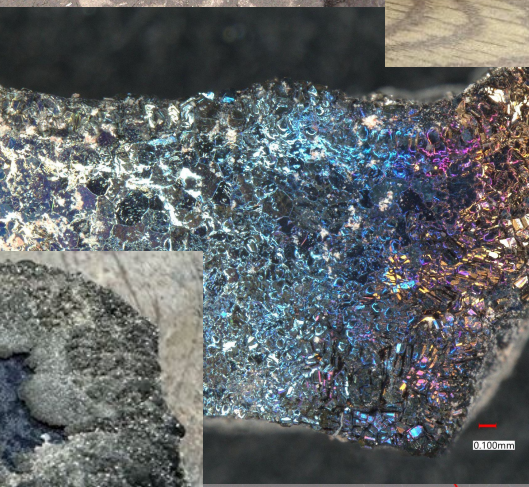
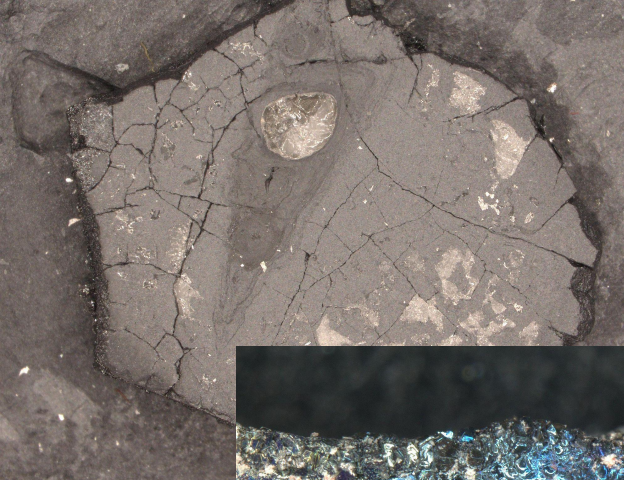
- In excess it changes the Ta to the color of Yttrium
- Diminishes the quality of the Susceptor

Carbon and Tantalum

- Turns the Ta gold colored, the TaC is very hard and appears to still be a good susceptor

Quartz? and Tantalum (2000C) (does not need to be in contact)

- Forms Geodes



Summary

If we want to go above 2200C it needs to be C for our susceptor and the C and the YSZ can not be touching

Try radiative next